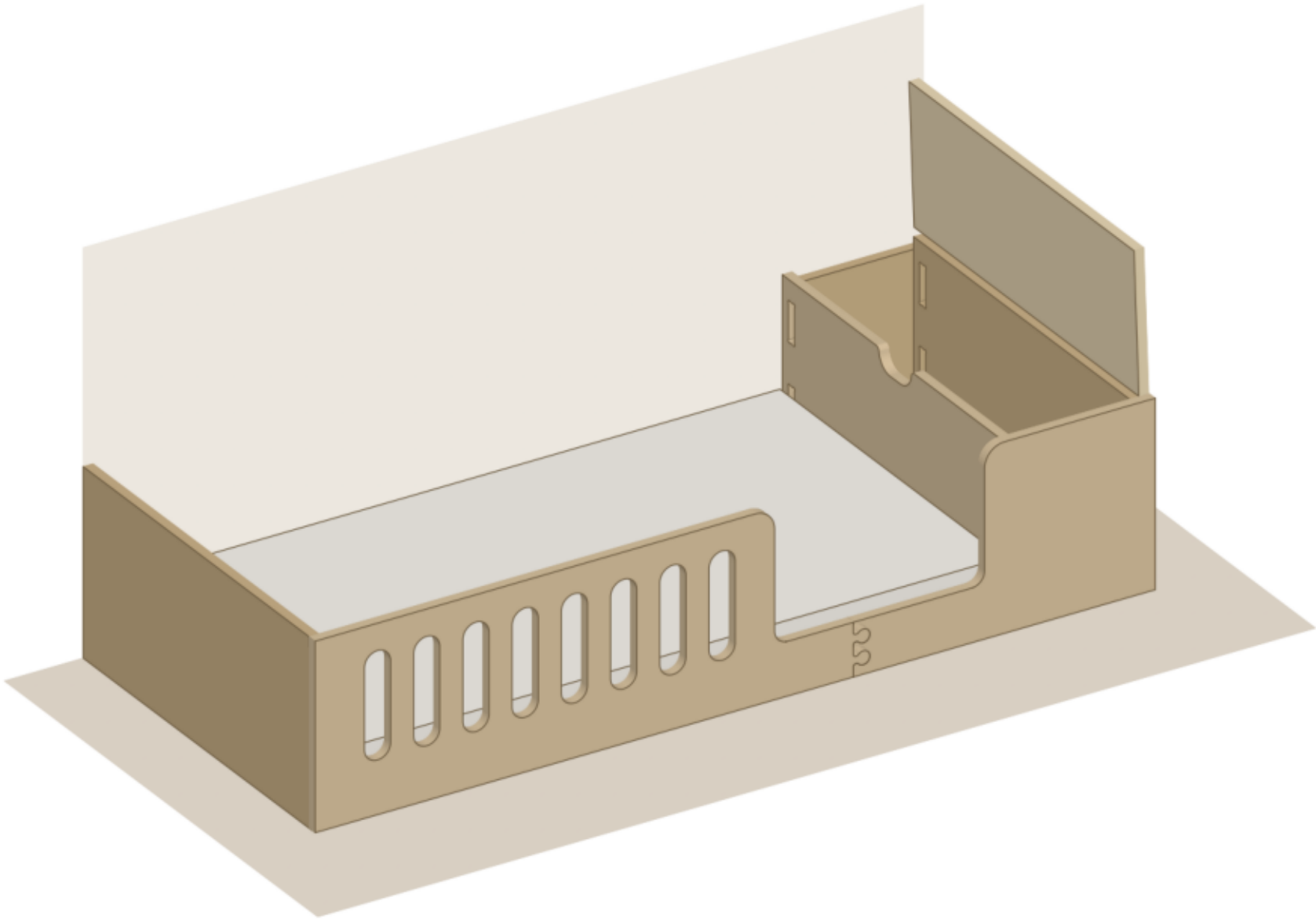
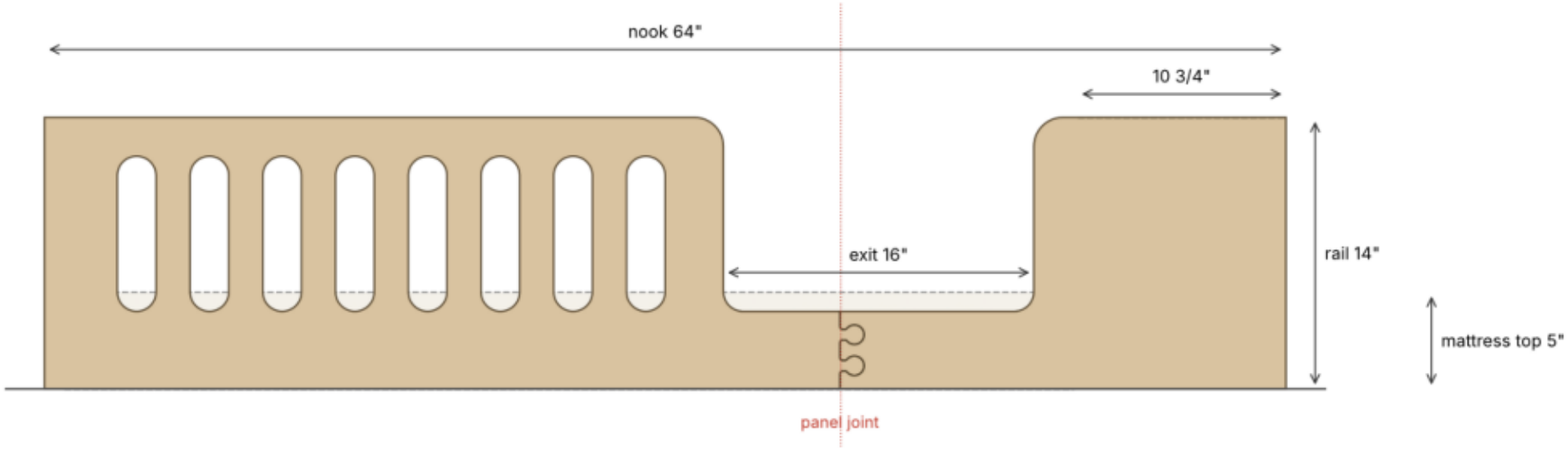
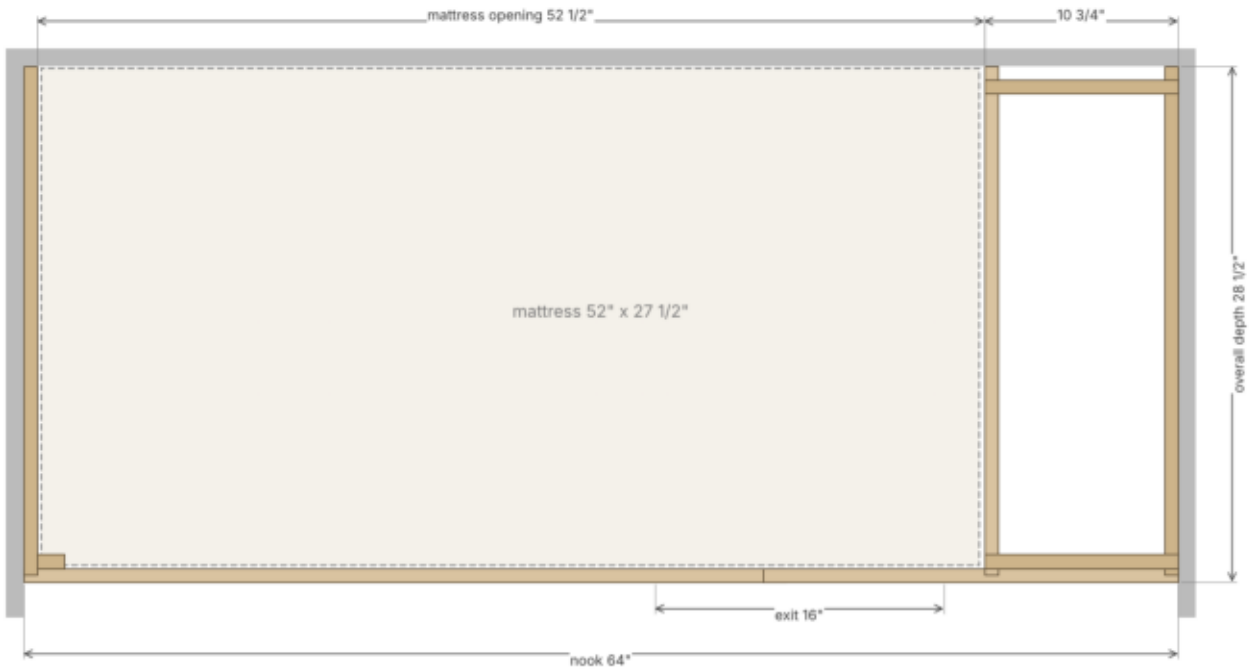
Floor bed, side exit, box at the foot



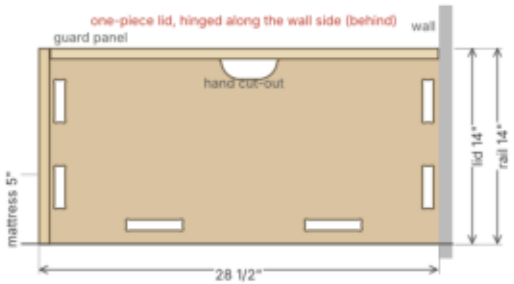
Floor bed, side exit, box at the foot: front view



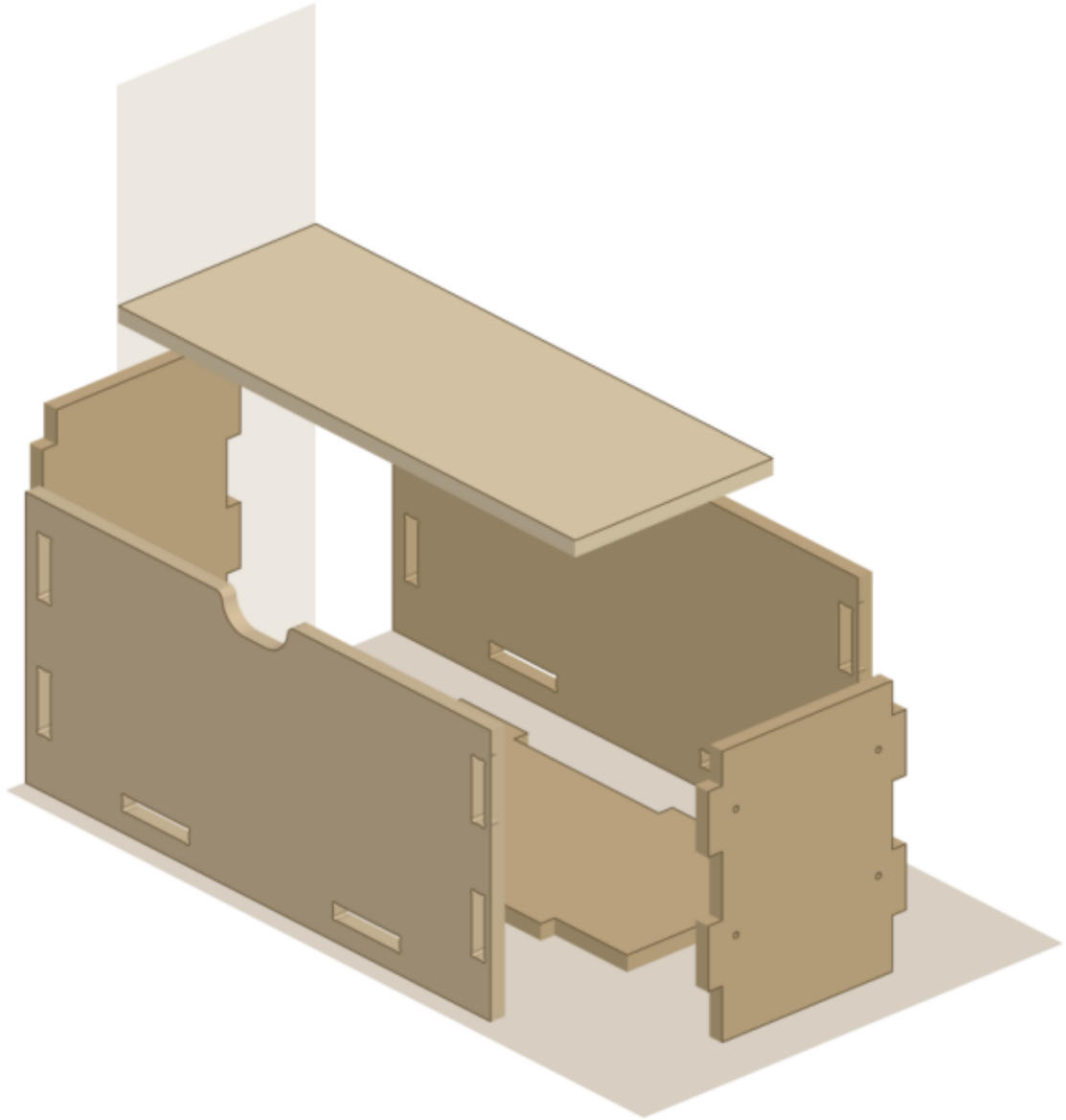
Plan (front of bed at the bottom)



Box seen from the mattress

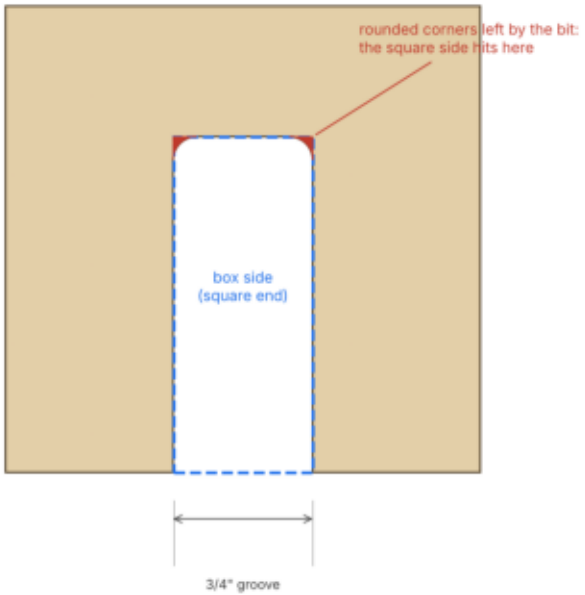


Foot box, exploded (tabs pass through the sides)

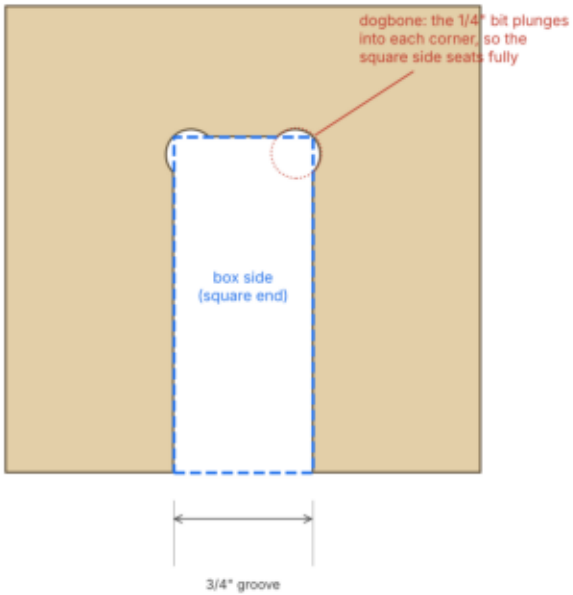


Dogbones: how a square part fits a CNC-cut groove (looking at the back of the front panel)

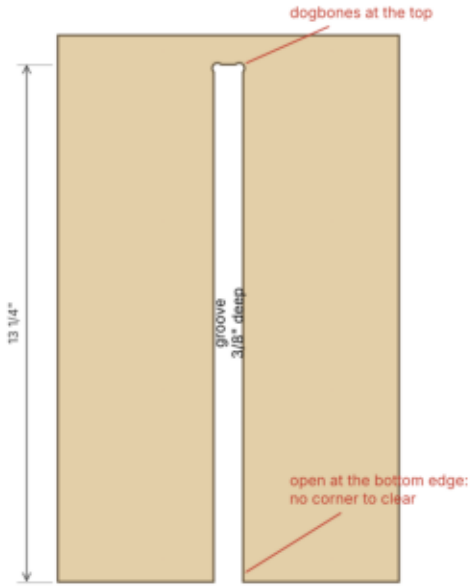
Without dogbones (red = wood that blocks the square corner)



With dogbones

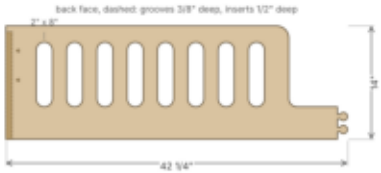


Back of the front panel, one groove

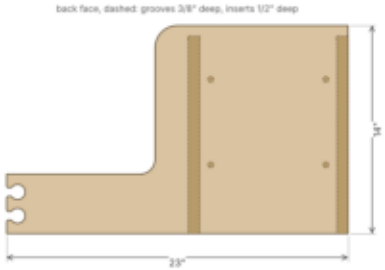


Floor bed, side exit, box at the foot: part drawings (inches, 3/4" plywood)

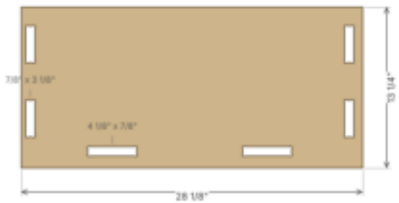
Front panel A (qty 1)
knob half



Front panel B (qty 1)
socket half



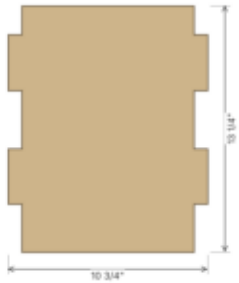
Box side (wall) (qty 1)
against the nook wall, 1 per box



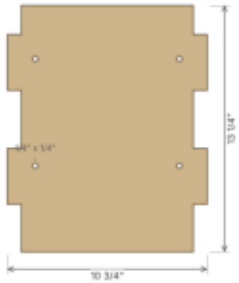
Box side (mattress) (qty 1)
faces the mattress; hand cut-out to open the lid



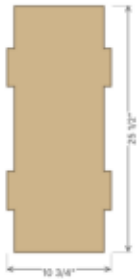
Box back (qty 1)
screw through into wall studs



Box front (qty 1)
against the back of the front panel; 4 bolts into it



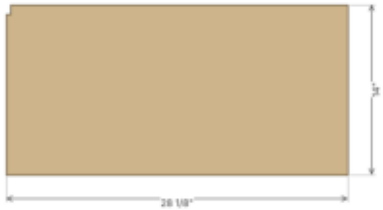
Box floor (qty 1)
sits 1" off the floor



Box lid (qty 1)
piano hinge along the long wall-side edge



Head wall liner (qty 1)
screw to the head wall studs; runs into the panel groove

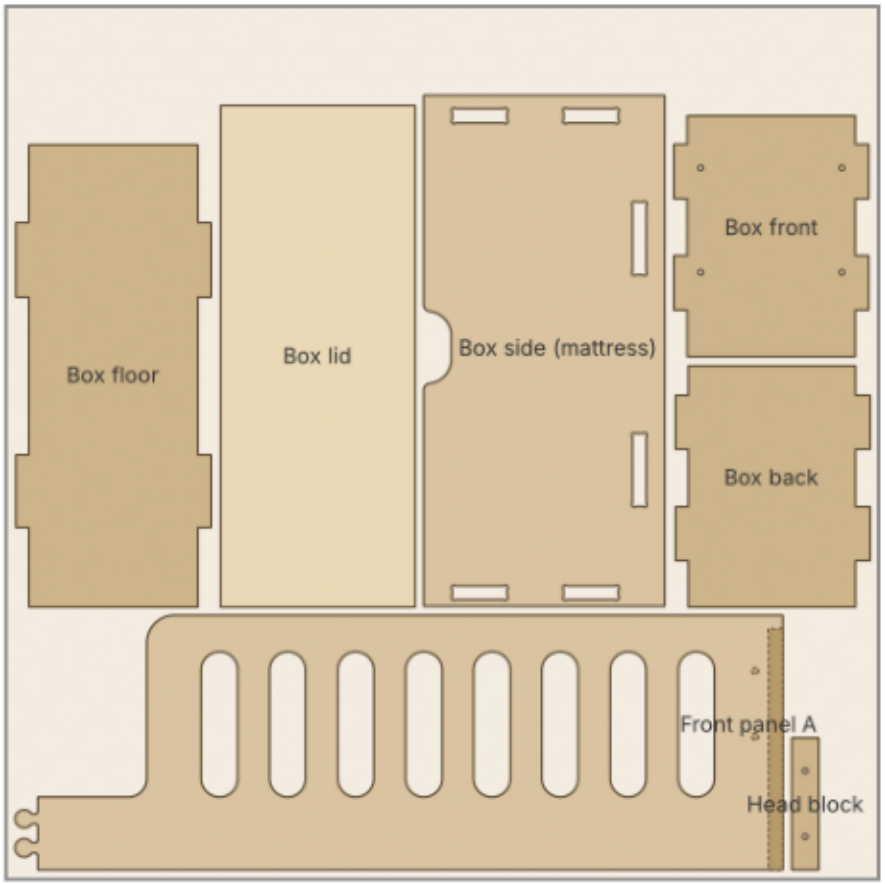


Head block (qty 1)
above the mattress; glue + 2 screws to the liner, 2 bolts into the panel

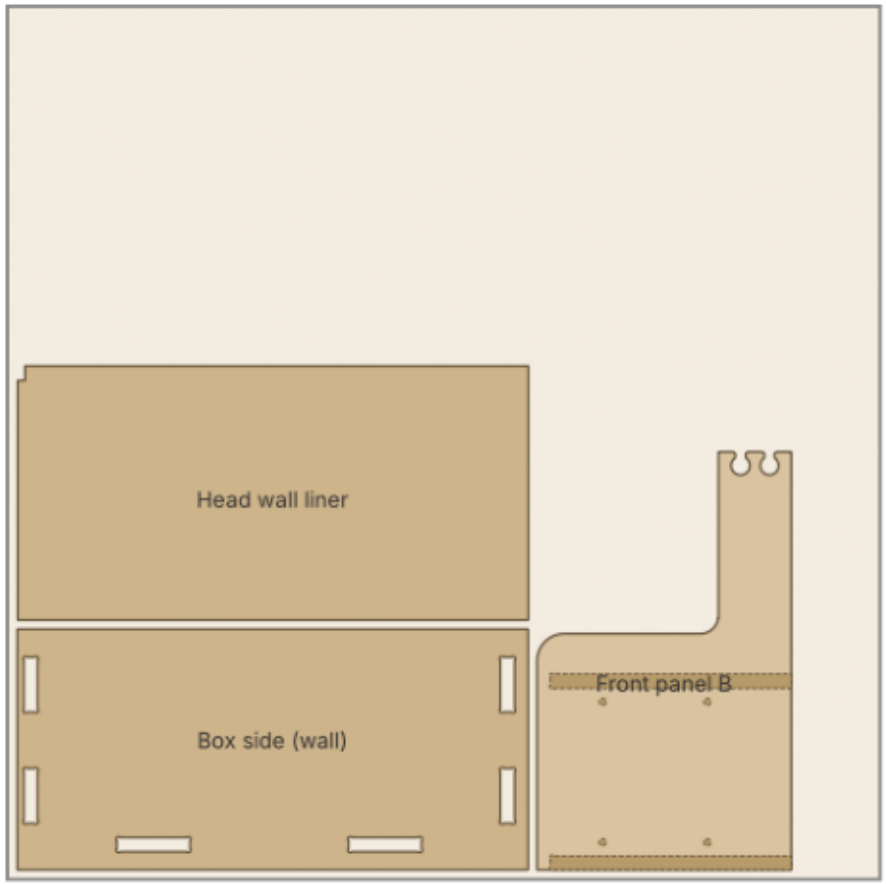


Floor bed, side exit, box at the foot: nesting on 48" square panels

4'x4' panel 1



4'x4' panel 2



Floor bed, side exit, box at the foot: cut list, hardware, assembly

Cut list

Part	Qty	Size (W x H)	Notes
Front panel A	1	42 1/4" x 14"	knob half
Front panel B	1	23" x 14"	socket half
Box side (wall)	1	28 1/8" x 13 1/4"	against the nook wall, 1 per box
Box side (mattress)	1	28 1/8" x 13 1/4"	faces the mattress; hand cut-out to open the lid
Box back	1	10 3/4" x 13 1/4"	screw through into wall studs
Box front	1	10 3/4" x 13 1/4"	against the back of the front panel; 4 bolts into it
Box floor	1	10 3/4" x 25 1/2"	sits 1" off the floor
Box lid	1	10 3/4" x 27 5/8"	piano hinge along the long wall-side edge
Head wall liner	1	28 1/8" x 14"	screw to the head wall studs; runs into the panel groove
Head block	1	1 1/2" x 7 1/4"	above the mattress; glue + 2 screws to the liner, 2 bolts into the panel

Key dimensions

Panels: 2 pieces of 4'x4' (1 x 4'x8' sheet(s) of 3/4" plywood, each cut in half)
Heights from floor: mattress top 5", rail top 14", box lids 14", exit sill 4"
Exit opening: 35" to 51" from the head wall; panel halves join at 41"
Foot box: 10 3/4" wide outside, 9 1/4" inside, 27 3/4" deep; no head box (mattress against a liner on the head wall)

Hardware

- Wood glue (PVA)
- 1-1/4" wood screws x 14 (box joinery + head block)
- 1/4-20 threaded inserts for wood x 6, no longer than 3/8" (front panel)
- 1/4-20 x 1" bolts + washers x 6 (box fronts and head block to front panel)
- 2-1/2" screws x 5 into studs (box back and head liner)
- Piano hinge 1-1/2" open width, 1 x 24" length(s) (1 per lid)
- Soft-close lid stays x 1 (light-duty toy-box type)
- Shims as needed

Assembly

1. Cut all parts (sheets/panel_*.dxf). The two front-panel pieces are mirrored in the files: cut them BACK FACE UP so the POCKET layers (box-side grooves 3/8" deep, insert holes 1/2" deep) land on the back. Profile CUT_INSIDE before CUT_OUTSIDE, use tabs/onion skin on small parts, and dry-fit the puzzle joint and one groove on scrap first; adjust knob_clearance / tab_clearance if needed.
2. Ease every exposed edge with a 1/4" roundover (rail top, slots, exit opening, lid edges and the